THE NORMALIZATION IS THE COPRODUCT OF THE NORMALIZATIONS OF THE REDUCED IRREDUCIBLE COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 14587-14594 and proof span 14596-14603 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof offers two alternative deductions, as the author wrote them; both are preserved.

Editorial note. The statement and proof use the author's notation for the reduced irreducible components and their normalizations; it is preserved.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The normalization is unchanged by passing to the reduction of the base, cited from the same chapter. Cited at proof lines 14596-14597. Allows the base to be assumed reduced.
- The local description of the normalization by integral closures, and the structural statement that the normalization is universal for factorizations through locally birational integral morphisms, both cited from the same chapter. Cited at proof lines 14598-14603. Give two alternative routes to identify the coproduct as the normalization.
- For a reduced scheme with locally finitely many irreducible components the coproduct of the components is integral and locally birational over the base. Used at proof lines 14601-14603. Supplies the hypothesis under which the structural statement applies.

Source reference. Original lemma 035P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4136–4136. Same-chapter reference used in the retained author text at source line(s) 14491, 14535, 14570, 14599, 14678, 14700, 15083. It is supplied only as a navigation pointer so that the retained text compiles; the

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Source reference. Original lemma 035Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4137–4137. Same-chapter reference used in the retained author text at source line(s) 14600, 14613, 14657, 14662, 14719. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035O: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4135–4135. Same-chapter reference used in the retained author text at source line(s) 14534, 14561, 14597. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme such that every quasi-compact open has finitely many irreducible components. Let $Z_i \subset X$, $i \in I$ be the irreducible components of X endowed with the reduced induced structure. Let $Z_i^\nu \to Z_i$ be the normalization. Then $\coprod_{i \in I} Z_i^\nu \to X$ is the normalization of X.*

Proof. We may assume X is reduced, see Lemma 035O. Then the lemma follows either from the local description in Lemma 035P or from Lemma 035Q part (3) because $\coprod Z_i \to X$ is integral and locally birational (as X is reduced and has locally finitely many irreducible components). □